进度汇报

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A. Jungo, F. Balsiger, and M. Reyes, “Analyzing the Quality and Challenges of Uncertainty Estimations for Brain Tumor Segmentation,” Front. Neurosci., vol. 14, p. 282, Apr. 2020.

出发点

- 自动分割算法能够很好的分割肿瘤，但是仍然不能满足实际中对于分割结果鲁棒的要求。
- 不确定性估计可以比较好的解决这一点。

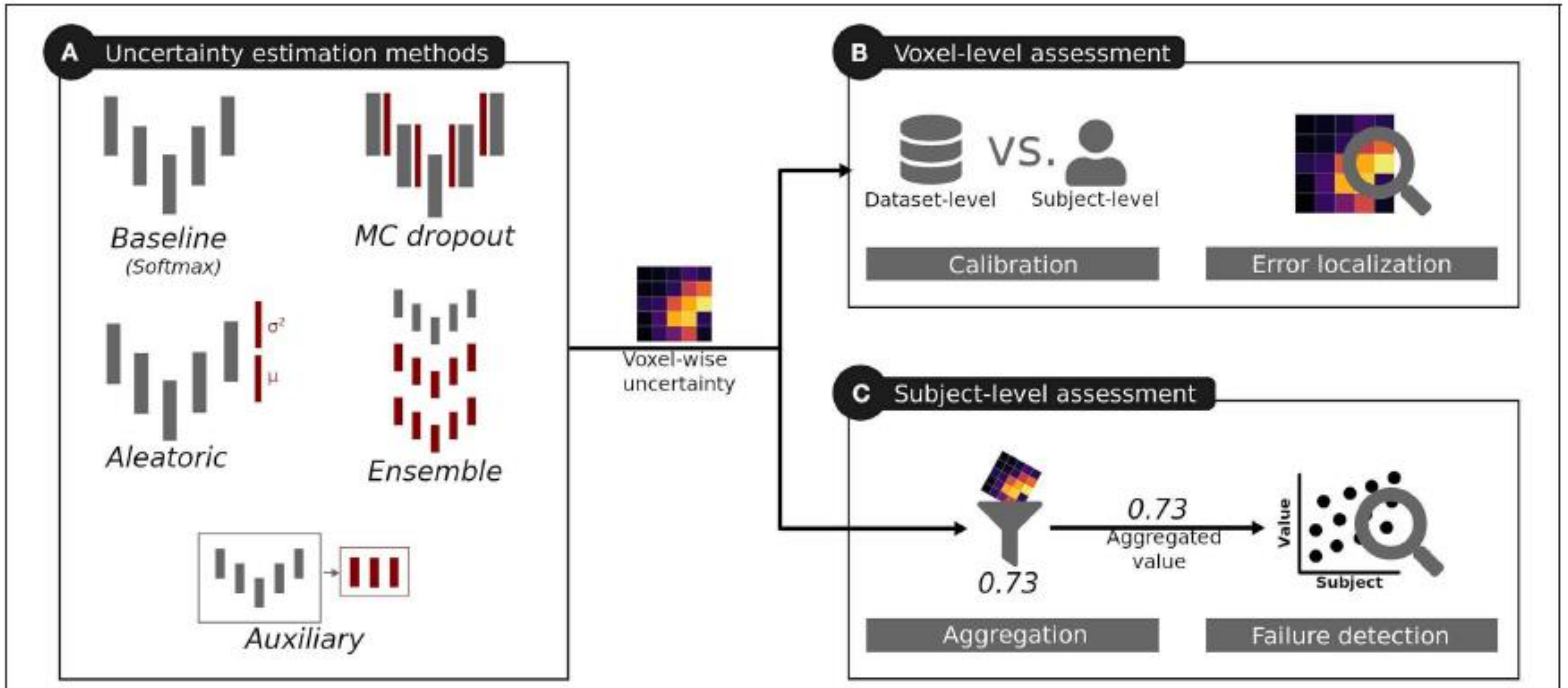

FIGURE 1 | Overview of the analysis performed for the uncertainties produced by different uncertainty estimation methods. The red color indicates additions introduced by these methods with respect to the baseline.

baseline

2.3.1. Baseline: Softmax/Sigmoid Uncertainty

Although the softmax/sigmoid output is arguably a probability measure (Gal and Ghahramani, 2016), it is implicitly produced by classification networks. Therefore, we considered it as a reference comparison and named it *baseline*. We used the normalized entropy

$$\mathcal{H} = -\left[p_r log\left(p_r\right) + (1 - p_r) log\left(1 - p_r\right)\right] \frac{1}{log(2)} \in [0, 1] \quad (1)$$

MC Dropout

$$p_r = \frac{1}{T}\sum_{t=1}^{T} p_{r,t} \,,$$

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MC Dropout

As shown by Gal and Ghahramani (2016), test-time dropout can be interpreted as an approximation of a Bayesian neural network. If applied during test time, dropout creates stochastic network samples that can be viewed as Monte-Carlo samples drawn from the posterior distribution of the network's weights. The foreground probability p_r of the tumor region r can be obtained by

$$p_r = \frac{1}{T} \sum_{t=1}^{T} p_{r,t} \,,$$

where T is the number of samples. We used the normalized entropy (Equation 1) as a measure of uncertainty.

Ensembles

2.3.4. Ensembles

Ensembles of neural networks are typically used when performance is highly relevant, e.g., for the BraTS challenge (Kamnitsas et al., 2017), but they can also be used to quantify uncertainties (Lakshminarayanan et al., 2017). Our ensemble consists of $K = 10$ models that share the same architecture (see section 2.2) but differ in training to enforce variability. We trained each model k on alternating $K - 1$ folds of the training dataset (as in k-fold cross-validation, resulting in 90 instead of 100 training subjects. We obtained the foreground probability p_r for each tumor region r by the average

$$p_r = \frac{1}{K} \sum_{k=1}^{K} p_{k,r}$$

of all models. As an uncertainty measure we used the normalized entropy (Equation 1).

Auxiliary Networks

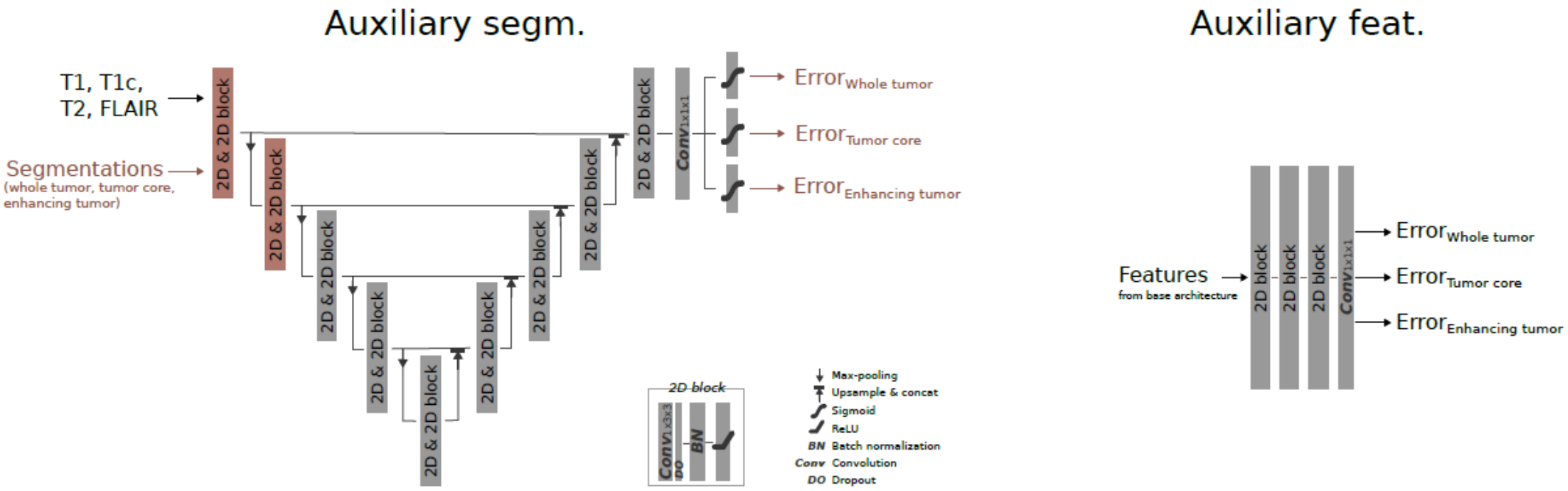


Figure S2. Structure of the *auxiliary segm.* and *auxiliary feat.* architectures. The modifications of the *auxiliary segm.* architecture compared to the base architecture are shown in brown.

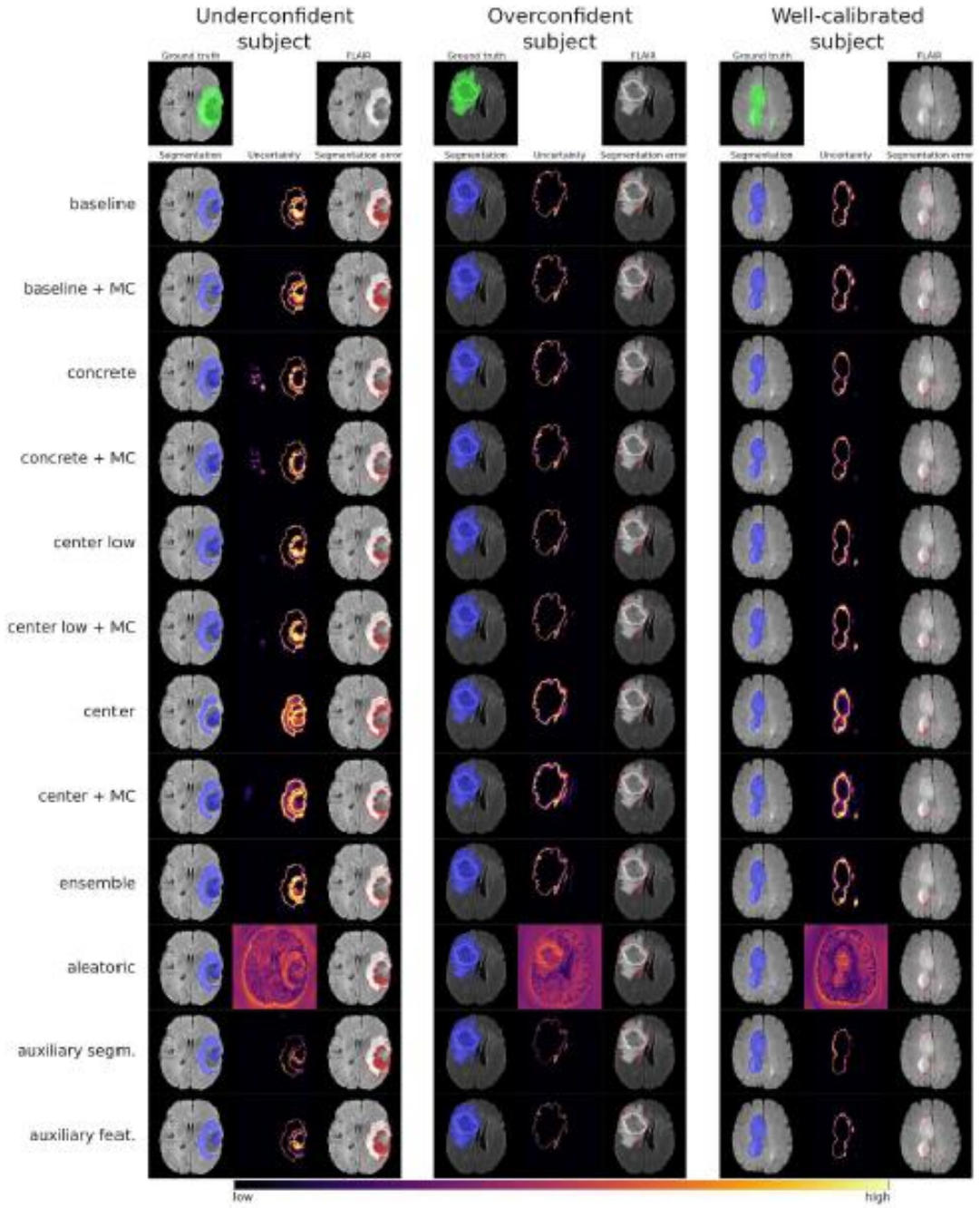
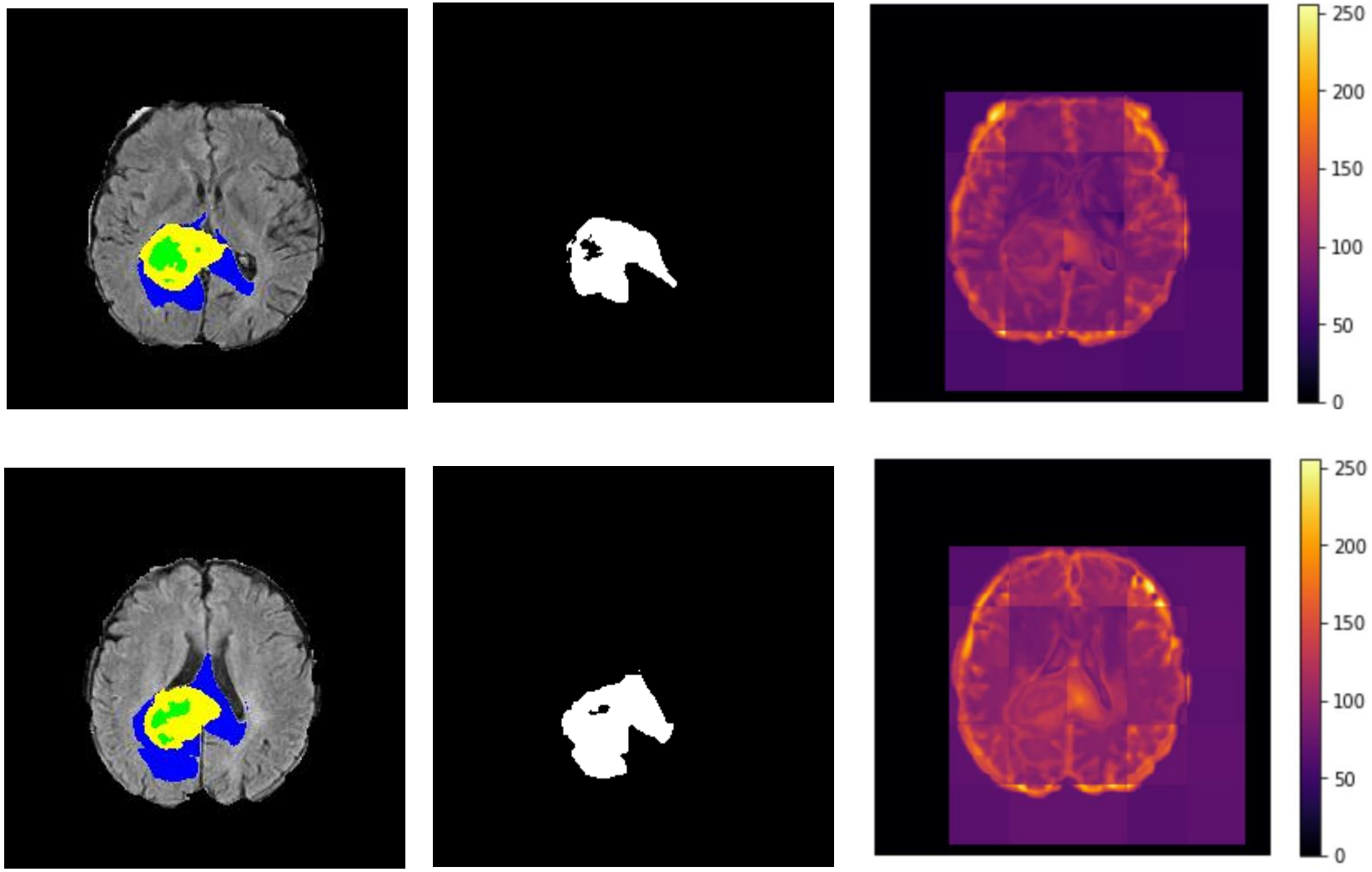


FIGURE 3 | Visual examples of the whole tumor uncertainty produced by the different uncertainty estimation methods. The columns correspond to underconfident, overconfident, and well-calibrated subjects (same as in **Figure 2**).

我的实验结果



Thank You!